DUBLIN CITY UNIVERSITY

SEMESTER TWO EXAMINATIONS 2020/21

MODULE:	Linear algebra MS200
QUALIFICATIONS:	CASE: BSc in Computer Applications (Software Engineering) SE: BSc Science Education PEM: BSc in Physical Education with Maths
YEAR OF STUDY:	2
EXAMINERS:	Dr. Abraham Harte (Ext: 5669) Dr. Stephen O'Sullivan Dr. Hitesh Tewari
TIME ALLOWED:	2 hours
INSTRUCTIONS:	Attempt all questions. Try to organize your reasoning as much as possible in your uploaded work. Marks will be awarded for clarity and evidence of understanding. Marks will not be awarded for irrelevant or contradictory statements which make no significant progress towards a solution. Note that questions do not carry equal marks.
REQUIREMENTS:	Any amount of printed information may be used. This includes notes, books, etc. A calculator is allowed as well.

QUESTION 1 [TOTAL MARKS: 18]

Let

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ -1 \end{pmatrix}, \qquad \mathbf{y} = \begin{pmatrix} 3 \\ -1 \\ -3 \\ 1 \end{pmatrix}.$$

(a) Find $|\mathbf{x}|$, $|\mathbf{y}|$, and the angle between $\mathbf{x}$ and $\mathbf{y}$.

[6 marks]

(b) Given a straight line which passes through the origin and is tangent to $\mathbf{x}$, find the distance between that line and the point which is separated from the origin by $\mathbf{y}$.

[6 marks]

(c) Suppose that part (b) is asked again, but with $\mathbf{y}$ replaced by a different point $\mathbf{z}$. If the distance from $\mathbf{z}$ to the given line is found to be equal to $|\mathbf{z}|$, what does that imply about the angle between $\mathbf{x}$ and $\mathbf{z}$?

[6 marks]

QUESTION 2 [TOTAL MARKS: 18]

Consider the linear functions $\mathbf{f} : \mathbb{R}^3 \to \mathbb{R}^3$ and $\mathbf{g} : \mathbb{R}^2 \to \mathbb{R}^3$ given by

$$\mathbf{f}(\mathbf{x}) = \begin{pmatrix} 2x_1 - x_3 \\ x_2 \\ x_1 + x_2 \end{pmatrix}, \qquad \mathbf{g}(\mathbf{y}) = \begin{pmatrix} y_1 \\ -2y_1 \\ y_2 \end{pmatrix}.$$

(a) Find matrices A and B such that $\mathbf{f}(\mathbf{x}) = A\mathbf{x}$ and $\mathbf{g}(\mathbf{y}) = B\mathbf{y}$ for all $\mathbf{x} \in \mathbb{R}^3$ and all $\mathbf{y} \in \mathbb{R}^2$.

[6 marks]

(b) Indicate which of the following functions are well-defined: $\mathbf{f} \circ (\mathbf{f} \circ \mathbf{g})$, $\mathbf{g} \circ \mathbf{f}$, $\mathbf{g} \circ \mathbf{g}$, $\mathbf{f} + \mathbf{g}$.

[4 marks]

(c) Find the matrices associated with any function(s) in part (b) which are well-defined.

[8 marks]

QUESTION 3 [TOTAL MARKS: 13]

Let

$$A = \begin{pmatrix} 1 & 3 & -2 \\ 2 & 1 & 1 \end{pmatrix}$$

(a) Use Gauss-Jordan elimination to put A into reduced row echelon form.

[7 marks]

(b) Write down a linear system involving two equations and two unknowns for which A is the associated augmented matrix.

[6 marks]

QUESTION 4 [TOTAL MARKS: 14]

Suppose that the components of a 3×3 matrix A are given by

$$(A)_{ij} = i - j$$

for all $i, j \in \{1, 2, 3\}$.

(a) Comment on whether is A square, symmetric, antisymmetric, or the identity matrix. Note that more than one of these labels may apply.

[4 marks]

(b) Find the characteristic equation for A. Also solve that equation, finding all three eigenvalues. Note that two of these eigenvalues are imaginary.

[10 marks]

QUESTION 5 [TOTAL MARKS: 15]

Suppose that a ball is dropped and its height is measured at n times, resulting in the data (t_1, h_1), (t_2, h_2), $\ldots$, (t_n, h_n). Constructing the $n \times 3$ matrix M with components $(M)_{ij} = t_i^{j-1}$, where the i in t_i^{j-1} denotes an index and the $j-1$ an exponent, that data is found to satisfy

$$M^T M = \begin{pmatrix} 50.0 & 51.0 & 68.7 \\ 51.0 & 68.7 & 104 \\ 68.7 & 104 & 168 \end{pmatrix}, \qquad M^T \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix} = -\begin{pmatrix} 142 \\ 264 \\ 461 \end{pmatrix}.$$

(a) The experimenter has lost their raw data and forgotten n. Reconstruct n from the above "processed data." Hint: It may be useful to think about the first column of M.

[7 marks]

(b) Use the least-squares technique to estimate the ball's acceleration g to two significant figures, which can be found by fitting to $h = a + bt - \frac{1}{2}gt^2$. You may use that

$$(M^T M)^{-1} = \begin{pmatrix} 0.195 & -0.386 & 0.159 \\ -0.386 & 0.998 & -0.460 \\ 0.159 & -0.460 & 0.225 \end{pmatrix}.$$

[8 marks]

QUESTION 6 [TOTAL MARKS: 22]

Consider a sequence $f_0, f_1, f_2, \ldots$ which satisfies the recurrence relation

$$f_n = \alpha f_{n-1} + \beta f_{n-2}$$

for all $n \geq 2$, where α and β are real constants.

(a) This recurrence relation can equivalently be written as $\mathbf{x}_n = R\mathbf{x}_{n-1}$ for all $n \geq 2$, where R is a matrix and

$$\mathbf{x}_n = \begin{pmatrix} f_n \\ f_{n-1} \end{pmatrix}.$$

Find R.

[6 marks]

(b) If $\alpha = 2$ and $\beta = 1$, it is possible to show that

$$R = P \begin{pmatrix} 1+\sqrt{2} & 0 \\ 0 & 1-\sqrt{2} \end{pmatrix} P^{-1},$$

where

$$P = \begin{pmatrix} 1+\sqrt{2} & 1-\sqrt{2} \\ 1 & 1 \end{pmatrix}.$$

You do not have to show this. Instead, use it to write down the eigenvalues and eigenvectors of R when $\alpha = 2$ and $\beta = 1$.

[8 marks]

(c) Assuming that the first component of $P^{-1}\mathbf{x}_1$ is nonzero, use the form for R which is given in part (b) to find $\lim_{n\to\infty}(f_n/f_{n-1})$ when $\alpha = 2$ and $\beta = 1$.

[8 marks]